

When Good Models Fail: A Sociotechnical Audit of Predictive Systems

DNSC 6330: Responsible Machine Learning The George Washington University, School of Business

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Part A: Conceptual Framework

A.1 Distribution Drift

A model trained on distribution $P_{\text{train}}(X, Y)$ is deployed into a world that draws new observations from $P_{\text{deploy}}(X, Y)$. Distribution drift occurs when $P_{\text{train}} \neq P_{\text{deploy}}$, silently eroding reliability without any change to the model itself. Three operationally distinct types of drift must be carefully distinguished because they require different remedies.

Covariate shift occurs when the marginal distribution of inputs changes while the conditional relationship between inputs and outcomes remains stable:

$$P_{\text{deploy}}(X) \neq P_{\text{train}}(X) \quad \text{but} \quad P(Y | X) \text{ is unchanged}$$

In a recidivism-prediction context, a policy change that diverts younger defendants away from the criminal justice system would alter the age distribution of the scoring population without altering the underlying relationship between prior arrests and future recidivism. The model has not become less accurate in principle - it is simply being applied to a shifted input population, and its predictions may be poorly calibrated for that new population. The appropriate remedy is importance weighting to re-weight training observations toward the deployment distribution.

Label shift (also called prior shift) occurs when the marginal distribution of outcomes changes while the class-conditional distribution of features given outcome remains stable:

$$P_{\text{deploy}}(Y) \neq P_{\text{train}}(Y) \quad \text{but} \quad P(X | Y) \text{ is unchanged}$$

If a city adopts an intervention program that reduces overall recidivism rates, the base rate $P(Y = 1)$ falls without changing what a person who does reoffend looks like. The model's score thresholds, calibrated to the old base rate, will now over-predict risk. The appropriate remedy is threshold recalibration rather than retraining.

Concept drift is the most severe form: the conditional relationship $P(Y | X)$ itself changes - the world's data-generating mechanism evolves. In credit scoring, the introduction of Buy Now Pay Later (BNPL) financial products altered what prior payment history predicts about future default; the learned relationship was not merely shifted but structurally different. Concept drift invalidates the model's core learned function and typically requires full retraining or online learning. It can silently degrade both performance and fairness simultaneously, because the model's learned reliance on proxy variables may become more or less discriminatory as the world changes.

The distinction between shift types matters operationally: misdiagnosing label shift as covariate shift leads to incorrect corrections that may amplify rather than reduce error. A monitoring stack must test for all three types independently. Input distribution tests (PSI, KS, MMD) are leading indicators - detectable before harm occurs. Output performance tests (rolling AUC windows, Page-Hinkley sequential change-point detection) are lagging indicators - detectable only after the model has already degraded in deployment.

Why performance without monitoring is not sufficient: A model that achieves strong held-out AUC on a static test set at training time provides no guarantee of reliability at month six or year three of deployment. Standard train/test evaluation cannot detect drift by design - both sets are drawn from the same historical distribution. Drift is an inherently post-deployment phenomenon.

Without a monitoring stack that continuously tracks input distributions and output performance across time and subgroups, degradation is invisible until it has already caused harm at scale. Responsible deployment requires treating monitoring as a first-class engineering requirement, not an optional enhancement appended after deployment.

A.2 Generalization and the Generalization Gap

Generalization refers to a model's ability to perform on unseen but distributionally similar data. Formally, the generalization gap is the difference between true risk on the deployment distribution and empirical risk on the training set:

$$G_{\text{gap}}(\hat{h}) = R_{\text{true}}(\hat{h}) - R_{\text{emp}}(\hat{h}) = \mathbb{E}_{(x,y) \sim P_{\text{test}}}[\ell(\hat{h}(x), y)] - \mathbb{E}_{(x,y) \sim P_{\text{train}}}[\ell(\hat{h}(x), y)]$$

A large positive gap signals **overfitting**: the model has memorized training-specific patterns - noise, dataset artifacts, or confounds - rather than learning stable signal. A high-variance model that performs well in validation but fails in production creates audit-trail gaps, because its excellent validation metrics provide false assurance to decision-makers and regulators who rely on them.

Overfitting is a governance problem, not merely a statistical one. A model deployed to inform liberty-affecting decisions (bail, parole, child welfare screening) must be stable across the full range of individuals it will actually encounter in deployment, not merely well-calibrated to the historical training cases from which it was built.

Shortcut learning (Geirhos et al., 2020) is a related failure mode: the model correctly predicts labels in training by exploiting features that happen to correlate with the outcome in the training sample but fail to maintain that correlation in the deployment context. Three levels of shortcut are operationally relevant:

1. **Dataset shortcuts:** Label leaks through metadata, scanner identifiers, or demographic proxies.
2. **Task shortcuts:** An easier proxy task is substituted for the harder true task the model was intended to learn.
3. **Deployment shortcuts:** context cues present during training are absent in the target deployment domain.

Standard cross-validation cannot detect shortcuts when train and test share the same confound - the confound appears as signal in both splits. This is why high held-out accuracy is a necessary but not sufficient condition for responsible deployment.

A.3 Spurious Correlations

A spurious correlation is a statistical association in the training data between a feature X_j and the label Y that does not reflect a causal or stable relationship, but rather a confound specific to the training sample or collection context:

$$\text{Cov}_{\text{train}}(X_j, Y) \neq 0 \quad \text{but} \quad \text{Cov}_{\text{deploy}}(X_j, Y) \approx 0$$

Classic examples span domains. A chest X-ray model learned to predict pneumonia from scanner manufacturer ID rather than disease signal (Zech et al., 2018). A CNN tested for tank detection learned to distinguish sunny from cloudy skies, not tanks. In natural language processing, a sentiment model relied on the position of the token “not” rather than semantic content.

The credit-scoring analog is direct: a model trained 2015–2019 associates stable employment with creditworthiness during an economic expansion. COVID-19 breaks this association because furloughed workers with excellent credit are suddenly high-risk by traditional scoring logic. The correlation was real in training and absent in deployment - making any credit decision based on it after 2020 both inaccurate and potentially discriminatory.

Why spurious correlations are a governance problem: Standard cross-validation cannot detect them if train and test share the same confound. Spurious correlations can encode protected attributes through proxy discrimination - a model may learn race not from an explicit race field but from features like zip code or prior arrest count that are themselves products of racially discriminatory enforcement patterns. A high-accuracy model may be making correct predictions for entirely wrong reasons, creating legal exposure that aggregate accuracy metrics will not reveal.

Detection strategy: Feature-importance audits using SHAP, combined with domain-expert review of features that should not be causally predictive, are the standard detection approach. Counterfactual perturbation testing - swapping the value of a protected attribute while holding all other features constant - provides a direct measurement of residual influence.

A.4 Robustness

A model is **robust** if its performance degrades gracefully - not catastrophically - when inputs deviate from the training distribution, whether through noise, realistic perturbation, distribution shift, or adversarial manipulation. Robustness and generalization are related but distinct: a model can generalize well on a held-out test set drawn from the same distribution as training, and still fail catastrophically when the deployment population shifts in composition or behavior. Responsible ML requires evidence of both properties.

Standard holdout and cross-validation evaluate average performance on the training distribution. They do not measure:

- Behavior under worst-case subpopulations (slice-level failure)
- Sensitivity to small but realistic perturbations (stress testing)
- Performance under conditions not represented in training (out-of-distribution inputs)
- Failure modes at deployment boundaries

Four robustness evaluation techniques are relevant to the audit:

1. **Slice-based evaluation** - compute performance metrics separately for meaningful subsets $S_k = \{(x, y) \in D_{\text{eval}}: \phi_k(x) = \top\}$. Failure on a slice is a failure mode even if aggregate performance is strong.
2. **Scenario stress testing** - define macroeconomic or operational scenarios (e.g., unemployment rises 3 pp, property values fall 20%) and perturb model inputs accordingly.

3. **Individual Conditional Expectation (ICE) curves** - plot $\hat{f}(x_{-j}^{(i)}, x_j)$ as x_j varies across its range while all other features are held fixed, revealing heterogeneous response patterns across individuals.

4. **Sensitivity analysis** - compute the variance of predictions as a function of each feature: $V_j = \text{Var}_{x_j}[\mathbb{E}_{x_{-j}}[\hat{f}(X)]]$. A high V_j with low domain justification signals spurious reliance.

In regulated industries, robustness testing takes a formal regulatory form: the Federal Reserve's Dodd-Frank Act Stress Test (DFAST) requires banks to demonstrate model stability under macroeconomic shock scenarios. The FDA requires AI/ML software device developers to document performance under distribution shift. These requirements reflect the same principle: a model that performs well on average is not defensibly deployable in high-stakes settings without evidence of stability under stress.

Part B: Measurement → Governance Mapping

The following table operationalizes the conceptual framework from Part A. Each row links a quantifiable risk type to the specific metric that detects it, the failure mode the metric reveals when triggered, and the governance policy action the detected failure requires. The numeric thresholds are derived from established industry practice (PSI conventions) and the Lecture 04 framework.

Risk Type	Metric	What It Detects	Failure Mode	Policy Implication
Covariate shift	Population Stability Index (PSI) on input features	Whether the marginal distribution of each input feature has shifted between training and deployment	Model is being applied to a population it was not validated on	PSI < 0.10: stable, no action required. PSI 0.10–0.25: flag for enhanced monitoring and document in MRM log. PSI > 0.25: mandatory model review; suspend automated decisions pending validation.
Covariate shift	Kolmogorov–Smirnov (KS) two-sample test on input features	Nonparametric test of whether two univariate distributions are statistically distinguishable	Input distribution shifted in ways PSI may underweight in the tails	$p < 0.05$: escalate to model validation team; retain audit record of test date, feature, and D -statistic.
Covariate shift (multivariate)	Maximum Mean Discrepancy (MMD ²) in encoded feature space	Joint shift in the full feature representation including interaction structure	Marginal PSI and KS both pass, but the joint feature space has drifted - univariate tests are insufficient	MMD ² significantly above zero: trigger multivariate review; do not rely on marginal tests alone

				when feature correlations are high.
Score distribution drift	PSI and KS on model output scores	Whether the model's predicted-probability distribution has shifted, independent of input drift	Model outputs are drifting despite stable inputs - signals possible concept drift or score miscalibration	Score PSI > 0.10: recalibrate decision thresholds. Score PSI > 0.25: initiate full retraining evaluation.
Generalization failure	Train vs. test AUC gap	Overfitting: model has memorized training-specific patterns rather than learning stable signal	Held-out performance is an unreliable predictor of deployment performance	AUC gap > 0.02: flag in MRM; restrict to lower-stakes decisions. Gap > 0.05: prohibit high-stakes deployment; require regularization audit or simpler model.
Generalization failure	Train vs. test log loss and Brier score gaps	Calibration degradation under train-test shift	Predicted probabilities are systematically biased in deployment	Document in MRM file; require recalibration on a fresh holdout window before next deployment refresh.
Spurious correlation	Permutation importance stability (train vs. test split)	Whether feature importance rankings are consistent across the train-test split	Model relies on features that lose explanatory power out of sample - shortcut learning signal	Top-5 feature rank changes > 2 positions between train and test splits: mandatory domain-expert review before deployment approval.
Spurious correlation	Counterfactual swap shift on protected attributes	Residual predictive signal carried by race, gender, or other legally sensitive attributes after controlling for all permitted features	Protected attribute is functioning as a predictive proxy - potential disparate impact liability	Mean absolute swap shift > 0.05 on any protected attribute: initiate LDA (Less Discriminatory Alternative) analysis before deployment approval.
Robustness - sensitivity	ICE curve range and global	How strongly predicted probability responds to	Model is over-reliant on one feature; small real-world	V_j high with low domain justification: document in MRM

	sensitivity index V_j	variation in a single feature across its full observed range	changes in that feature produce large individual decision changes	with explicit business justification. Range of mean score > 0.50 across feature range: require human review when that feature value is near inflection points.
Robustness - stress testing	Scenario stress on key features (perturb priors count by +2, +5, +10 units)	How many individuals flip from low-risk to high-risk classification under plausible operational perturbations	Model is fragile; small realistic changes produce large reclassification cascades	Share reclassified > 20% under a single-feature +2 perturbation: governance escalation; cap feature influence via regularization or decision ceiling.
Subgroup fairness	Slice-level FPR and FNR by race, gender, age	Differential error rates across protected-attribute subgroups	Model imposes systematically higher false-positive or false-negative rates on historically disadvantaged groups	FPR or FNR disparity ratio > 1.5× across groups: mandatory threshold audit per subgroup. Ratio > 3×: deployment requires independent external audit before proceeding.
Subgroup fairness	Slice-level AUC and Brier score	Differential predictive accuracy across subgroups, beyond error rates	Model is less accurate for some groups than others even when aggregate AUC is acceptable	Worst-slice AUC more than 0.05 below aggregate AUC: document as a known limitation; require a human override option for decisions affecting that subgroup.

Part C: Failure Scenario Analysis

C.1 Case Study: The COMPAS Recidivism Prediction System

The COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) risk-assessment tool presents the canonical illustration of a model with strong average performance concealing large subgroup error disparities. The audit numbers below are drawn from a full end-to-

end implementation of the Lecture 04 reliability audit pipeline applied to the ProPublica Broward County dataset, using Logistic Regression (LR) and Gradient-Boosted Tree (GBT) models trained on the same pipeline as in Lectures 01- 03.

Aggregate performance (test set):

Model	Test AUC	Test Accuracy	Test Log Loss	Test Brier Score
Logistic Regression (LR)	0.8329	0.7522	0.4934	0.1642
Gradient-Boosted Tree (GBT)	0.8309	0.7498	0.4993	0.1659

Examined at the aggregate level, both models exhibit respectable discriminative performance: AUC above 0.83, accuracy above 0.75. An analyst reviewing only these numbers might conclude the system is performing reliably and is ready for high-stakes deployment. The remainder of this section documents why that conclusion would be wrong.

C.2 Generalization Failure: GBT Overfitting

A train-versus-test performance comparison reveals that the GBT model shows modest but substantively important overfitting:

Model	Train AUC	Test AUC	AUC Gap	Train Accuracy	Test Accuracy	Accuracy Gap	Train Log Loss	Test Log Loss	Log Loss Gap
LR	0.8292	0.8329	-0.0037	0.7539	0.7522	+0.0017	0.5016	0.4934	-0.0082
GBT	0.8553	0.8309	+0.0245	0.7748	0.7498	+0.0250	0.4612	0.4993	+0.0381

The LR model generalizes near-perfectly - its test performance is essentially identical to training performance, indicating that the model has extracted stable signal rather than training-specific noise. The GBT's AUC gap of 0.024 and accuracy gap of 0.025 indicate that 200 estimators have begun to memorize training-specific patterns. This is not yet catastrophic, but it signals a model that is more fragile to distribution shift than its test-set metrics suggest. In a high-stakes deployment context with a shifting population, GBT's training-time advantage would be expected to erode more rapidly than LR's.

What the generalization gap misses: the average train-test gap does not reveal whether specific subgroups - whose distribution may differ more substantially from the training distribution - are experiencing larger degradation. A model can have a small aggregate generalization gap while overfitting severely for a minority subgroup.

C.3 Spurious Correlation: Race as a Residual Predictor

Counterfactual swap analysis - swapping each defendant's value on one attribute to its counterfactual while holding all other features fixed - measures the residual predictive signal carried by each attribute beyond the information in the permitted features:

Attribute Swapped	LR Mean Absolute Probability Shift	GBT Mean Absolute Probability Shift
Race (African-American ↔ Caucasian)	0.0806	0.0976
Gender (Female ↔ Male)	0.0251	0.0888
Crime type (Felony ↔ Misdemeanor)	0.0410	0.0699

Swapping race alone - with no other change to the feature vector - moves the average predicted probability by 8.1 percentage points under LR and 9.8 points under GBT. This is the largest shift of any tested attribute. Under LR, the race shift is 3.2× the gender shift, indicating that race is carrying disproportionate predictive weight relative to a legally protected attribute that itself already warrants LDA review.

GBT amplifies both race and gender sensitivity relative to LR - a governance concern for a model that is already marginally overfitting. The amplification pattern is consistent with a higher-capacity model that has more efficiently learned the spurious structure in the training data, not just the causal signal.

What the swap shift means: Race encodes residual predictive information even after controlling for prior arrest count, charge severity, and age.

What it misses: Mean Absolute Shift averages over the full test distribution; individual defendants at specific intersections of race, age, and prior count may experience much larger shifts than the mean suggests.

What actions it justifies: Trigger LDA analysis before deployment; flag collinear proxy features (especially `priors_count` in jurisdictions with racially disparate policing) for independent review.

The mechanism is not necessarily direct racial encoding. Race in the COMPAS context likely acts as a collinear proxy for neighborhood-level arrest patterns and policing intensity - features themselves shaped by decades of discriminatory enforcement practices. The model learns a stable statistical association that exists precisely because of historical structural inequity, not because race causes recidivism. A model trained on arrest data in a jurisdiction with racially biased policing will learn to predict further arrests partly by learning the policing pattern, not the underlying behavior - amplifying that pattern into the future.

C.4 Subgroup Disparity: Failure at the Slice Level

The aggregate AUC of 0.83 conceals severe disparities at the subgroup level. Per-slice evaluation reveals that certain groups bear a dramatically disproportionate share of the model's errors.

Race slices - Logistic Regression, test set:

Race Group	N (approx.)	Accuracy	FPR	FNR
African-American	~3,175	—	0.387	0.177
Caucasian	~2,103	—	0.109	0.563
Hispanic	~509	—	—	—
Other / Small groups	< 50 each	—	unreliable (n < 50)	unreliable (n < 50)

The false-positive rate for African-American defendants is 0.387: 38.7% of African-American defendants who did not reoffend were labeled high-risk. The corresponding rate for Caucasian defendants is 0.109 - a ratio of **3.55×**. African-American non-recidivists face more than three and a half times greater risk of erroneous high-risk classification than Caucasian non-recidivists.

The FNR pattern is the inverse and equally consequential: Caucasian defendants who did reoffend were labeled low-risk far more often (FNR 0.563) than African-American defendants who reoffend (FNR 0.177). The model systematically over-predicts risk for African Americans and under-predicts risk for Caucasians - a pattern entirely consistent with the Chouldechova impossibility theorem under unequal base rates (see Section C.5 below).

Age slices - Logistic Regression and GBT, test set:

Age Group	LR Accuracy	LR FPR	GBT Accuracy	GBT FPR
Less than 25	0.660	0.663	0.690	—
25–45	—	—	—	—
Greater than 45	—	—	—	—

Defendants under 25 face a false-positive rate of 0.663: nearly two in three young defendants who did not reoffend were labeled high-risk. This is by far the worst slice-level error rate in the audit. The age <25 slice also has the worst accuracy (0.660 for LR; 0.690 for GBT) - the model is both inaccurate and systematically biased against this group. The 25–45 and over-45 slices perform closer to aggregate accuracy, but the severe under-performance for the youngest defendants indicates that the model's learned relationship between prior arrests and recidivism generalizes poorly to individuals at the start of their criminal-history trajectory.

Gender slices - Logistic Regression, test set:

Female defendants show accuracy of 0.711 but an FNR of 0.469 - nearly half of female defendants who did reoffend were labeled low-risk by the model. The model systematically fails to identify genuine recidivism risk in women. This is a distinct harm pattern from the over-classification concentrated in African-American and young defendants: while those groups are over-criminalized by the model, women who pose genuine risk are systematically under-supervised, with potential public safety consequences alongside the gender-fairness concern.

C.5 Connection to the Chouldechova Impossibility Theorem

The FPR and FNR disparities documented in Section C.4 is not simply a design failure in COMPAS - they are the direct mathematical consequence of the impossibility theorem proven in Group Homework 3. Because African-American defendants have a higher base recidivism rate in the Broward County dataset:

$$\pi_{\text{Black}} = P(Y = 1 \mid A = \text{Black}) > P(Y = 1 \mid A = \text{White}) = \pi_{\text{White}}$$

no non-trivial, non-perfect classifier can simultaneously satisfy calibration (predictive parity) and equal FPR/FNR across groups. COMPAS scores were calibrated - a score of 7 carried similar empirical recidivism probability regardless of race - and the observed FPR and FNR disparities are the unavoidable arithmetic consequence of that calibration applied to groups with unequal base rates.

The disparity does not establish malicious intent. It establishes that a value choice - calibration over error-rate parity - was made implicitly, without documentation, and without disclosing the tradeoff to defendants, judges, or the public. The governance failure the theorem exposes is precisely the absence of that disclosure: the organization deployed a tool that satisfied one fairness criterion while violating two others, and presented the calibration result as if it were universal fairness.

C.6 Why Aggregate Metrics Concealed the Harm

Three compounding structural factors allowed the aggregate AUC of 0.83 to mask the severity of subgroup harm:

1. **Averaging over unequal group sizes.** African-American defendants constitute a larger share of the Broward County dataset than Caucasian defendants. A population-weighted average AUC will down-weight the Caucasian slice and partially absorb within-group error disparities into the aggregate statistic.
 2. **AUC is rank-based, not threshold-specific.** AUC measures discriminative capacity across all possible thresholds simultaneously. A model can have strong rank-ordering ability while assigning systematically biased predicted probabilities to one group - the bias appears in calibration and in threshold-specific metrics (FPR, FNR), not in AUC itself.
 3. **Intersectional harms are invisible to single-attribute slicing.** The analysis above examines race, gender, and age as independent dimensions. Young African-American women may face a compounded harm - over-classification from age and race combined with under-classification from gender - that is invisible in any single-attribute slice. Slice averages mask intersectional harms and likely understate the worst individual-level outcomes in the population.
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Part D: Less Discriminatory Alternative Reasoning

D.1 Framework

Under *Griggs v. Duke Power Co.* (1971) and subsequent EEOC guidance, an organization that identifies disparate impact - which the $3.55\times$ FPR disparity documented in Part C clearly establishes - must demonstrate that no Less Discriminatory Alternative (LDA) exists that meets the same business necessity with meaningfully lower discriminatory impact. The *Griggs* burden-shifting framework operates in three steps:

Step	Actor	Obligation
1	Plaintiff	Demonstrate a prima facie statistical disparity (e.g., FPR ratio $> 1.5\times$, or AIR < 0.80)
2	Defendant	Demonstrate business necessity - the practice is job-related and consistent with a legitimate operational need
3	Plaintiff	Demonstrate that a less discriminatory alternative exists that meets the same business necessity

The following two proposals constitute good-faith LDA candidates for the COMPAS context, each addressing a different point in the model pipeline.

D.2 LDA Proposal 1: Group-Specific Threshold Recalibration

Description. Rather than applying a single classification threshold across all defendants, calibrate a separate threshold τ_a for each demographic group a such that slice-level FPR is bounded at a common governance ceiling (e.g., $\text{FPR} \leq 0.20$ for all groups).

Mechanism. The predicted probability scores produced by the existing LR or GBT model remain unchanged - no retraining is required. Only the decision rule (the mapping from score to binary classification) is group-stratified. For groups with high base rates whose aggregate-threshold FPR exceeds the ceiling, the group-specific threshold is raised, reducing false-positive classifications. For groups already meeting the FPR ceiling, the threshold may be held constant.

What this means: Directly targets the metric at which disparity is most severe.

What it misses: Does not address the upstream source of the disparity - race continues to carry residual predictive signal in the model's scores. A future population shift could re-open the disparity even with recalibrated thresholds.

What actions it justifies: Deploy as an immediate remediation step while Proposal 2 (feature revision) is evaluated.

Dimension	Benefit	Cost
FPR disparity	Directly reduces $3.55\times$ disparity; all groups held to common FPR ceiling	None in terms of the metric targeted

FNR disparity	—	Raising threshold for high-base-rate groups also raises FNR in those groups, creating a new and opposite disparity: fewer true recidivists are identified, which may be framed as under-supervision
Calibration	Scores remain calibrated; only the decision boundary moves	Decision boundary is now explicitly race-aware, which may face legal challenge in jurisdictions prohibiting race-conscious decisions
Predictive utility	Aggregate accuracy impact is modest	Operational stakeholders may resist any reduction in overall predictive precision
Legal defensibility	Demonstrably reduces disparate impact; directly satisfies the Step 3 showing	Race-aware thresholding requires documented legal review; has been challenged in court in some contexts

Recommendation. Threshold recalibration is the lowest-cost, fastest-to-implement LDA. It does not require retraining, does not discard predictive information, and directly addresses the FPR metric at which disparity is most severe. The FNR tradeoff must be documented, disclosed, and reviewed by an independent governance committee before deployment. This proposal is complementary to, not a substitute for, Proposal 2.

D.3 LDA Proposal 2: Feature-Set Revision to Remove Collinear Race Proxies

Description. Retrain the model after removing or transforming features that function as collinear proxies for race - specifically, features whose mutual information with race remains high after controlling for the legitimate predictors. The counterfactual swap analysis in §C.3 provides direct evidence that race carries residual predictive signal; this proposal attempts to reduce that signal at its source in the feature encoding.

Mechanism. Conduct a systematic feature-level audit using mutual information analysis, SHAP dependence plots, and the counterfactual swap framework. Features whose swap sensitivity to race exceeds a governance threshold (e.g., features where deracializing the data reduces swap shift by more than 0.02) are candidates for removal or transformation. `priors_count`, while the strongest legitimate predictor of recidivism, may carry racial information in jurisdictions with racially disparate policing practices - it may be transformed to a categorical tier or combined with a geographic arrest-rate normalizer to reduce its proxy function while preserving its legitimate predictive content.

What this means: Addresses the root cause of racial signal in predictions, not just the decision rule.

What it misses: May not fully eliminate racial signal if remaining features are still jointly collinear with race; and some accuracy loss is expected if `priors_count` is modified.

What actions it justifies: Pursue in parallel with Proposal 1 as the longer-term structural fix; document each removed or transformed feature with an explicit causal justification for the MRM file.

Dimension	Benefit	Cost
Spurious correlation	Directly reduces the mechanism of racial proxy encoding at training time; smaller counterfactual swap shifts post-revision	May not fully eliminate racial signal if remaining features are still collinear with race
Predictive utility	No accuracy loss if removed features are pure proxies with no causal content	Estimated 2–5% AUC reduction if <code>priors_count</code> is partially modified; requires empirical validation
FPR disparity	Reduces racial signal in predictions; expected smaller FPR disparity in revised model	FPR reduction may be smaller than Proposal 1 alone; supplemental recalibration may still be required
Generalization	A model without spurious correlations should generalize more stably under population and policy shifts	Fewer high-information features may increase prediction variance in small-n slices (e.g., Asian, Native American groups)
Legal defensibility	Race-blind feature engineering is legally unambiguous; avoids challenges to race-aware decision rules entirely	Requires documented justification for each removed or transformed feature; may face stakeholder challenge if <code>priors_count</code> is modified

Recommendation. Feature-set revision is the structurally more durable LDA - it reduces disparate impact at the model level rather than at the decision-rule level - but carries higher implementation cost and a non-negligible accuracy tradeoff that must be validated empirically before deployment. It should be pursued in parallel with Proposal 1 rather than as a replacement. The two proposals address different points in the audit pipeline - upstream feature encoding and downstream decision threshold - and are complementary interventions.

D.4 Monitoring Requirements for Either LDA

Neither LDA is a one-time fix. The Chouldechova theorem guarantees that as long as base rates differ across groups, some fairness criterion will be violated. The governance obligation after implementing either proposal is to:

1. **Document the criterion chosen.** Record which fairness metric is being optimized (FPR ceiling in Proposal 1; reduced proxy encoding in Proposal 2), the numerical governance threshold, the rationale, and the date of the decision.
2. **Monitor on a rolling basis.** Track FPR, FNR, calibration, and AUC at the slice level on a defined cadence - quarterly in stable environments, monthly in high-volume or rapidly-shifting deployment contexts.

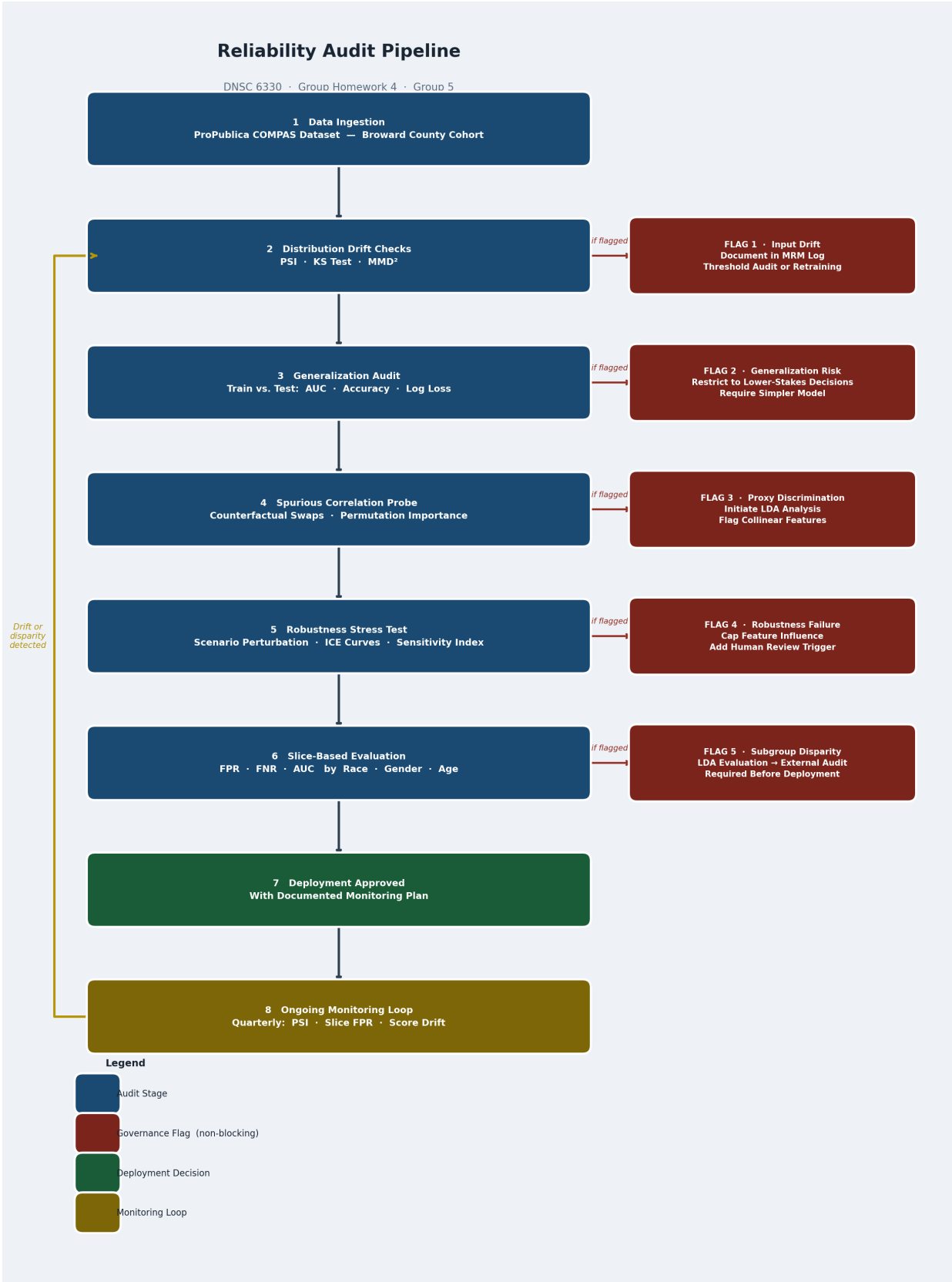
3. **Define a re-audit trigger.** If any monitored slice metric exceeds the governance threshold by more than a defined tolerance (e.g., FPR rises above the ceiling by more than 0.02), automatically re-initiate the LDA search.

4. **Commission independent audits annually.** External audit findings must be disclosed to relevant stakeholders - including, where applicable, the individuals whose decisions are being informed by the model.

Therefore, governance without monitoring is a report, not accountability.

Audit Pipeline Diagram

The diagram below illustrates the end-to-end reliability audit pipeline. The main pipeline (navy, left column) runs sequentially through eight stages. Each stage has a corresponding governance flag (dark red, right column) that is triggered when its threshold is breached - all flags are non-blocking checkpoints, meaning the audit continues through every stage before a final deployment decision is made. If drift or disparity is detected during the ongoing monitoring loop (amber), the pipeline restarts from the Distribution Drift Checks stage.



Audit Pipeline Diagram

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